

BI 028 Sam Gershman: Free Energy Principle & Human Machines

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you could build agents that adhere to those principles and many agents in that set would be what we consider smart and many of them would be stupid wait isn't that what we have already know just yeah right exactly so I really like what Sherlock Holmes said that to understand nature your mind must be as broad as nature that's how I feel like really the study of intelligence is that this is brain inspired hey everyone welcome to episode 28 of brain inspired I am Paul Middlebrooks I had been thinking of a way to talk about the free energy principle on the show and I was going to have a guest on a few weeks ago but we had to reschedule that for a later time but I stumbled on a recent paper by the person you just heard Sam Gershman who is a professor at Harvard University and runs the computational cognitive neuroscience lab there where they study learning memory decision-making you know cognition he's recently written about the free energy principle and how it relates to other bigger picture theories of how our brains function in particular the Bayesian brain hypothesis and the concept of predictive coding in general so we talked about just that during the first part of the show Sam also co-authored a paper a couple years ago now called building machines that learn and think like humans the paper breaks down some of the important ingredients that current deep learning networks are missing if we want to get to human-like intelligence so we talked about those concepts and how far along we've come since then so those are the two major themes during this show and of course we discuss much more like the limits of our own cognition and how that affects our ability to understand complicated processes in the world including our own intelligence and hang on until the end to because Sam gives some career advice that I love for those of you early in your career or even deep in your career but but trying to figure out your next course of action your next direction every new show I do I realize I have more and more questions to ask of the guests to try to bring it all together and keep you know some sort of running thread from the previous guests so I try to do that here

but of course I can never get to all the questions you can find out more about Sam and links to the stuff that we discuss in the show notes at brain inspired Co slash podcast slash 28 and you can also find how to support this ad free labor of love of a podcast but by finding and clicking on the red patreon button at brain inspired Co faire y'all see May I unnn and Justin did just that this past week thank you guys that is beautiful alright thank you for listening the hope that your projects are coming along smoothly and you're immersed in and loving whatever it is that you're pursuing enjoy the show all right Sam Gershman welcome and thanks for being on the show thank you so Sam you are a you run the computational cognitive neuroscience lab at Harvard and you do lots of things for instance you study how we learn you study the different systems that we use to learn and how they interact how our perceptions get transformed into these internal states in our brains and how those states can be used to guide our actions you study processes like exploration versus exploitation and plenty more today we are here to talk about two subjects in particular one the free energy principle which is a hot topic these days in in brain function and neuroscience in your recent paper what does the free energy principle tell us about the brain and two we're gonna talk a little bit about the concepts introduced in your paper from I guess two years ago now called building machines that learn and think like humans you ready mm-hmm the free energy principle what does the free energy principle tell us about the brain so in this paper you compare predictions of the free energy principle with the Bayesian brain hypothesis and how those two systems might fall out and compare to each other in this realm of what's called predictive coding now I'll just begin by reading the opening paragraph of a 2009 review by Karl Friston who has been the developer of the free energy principle just to get us kind of rolling on this topic here so he writes the free energy principle is a simple postulate with complicated implications it says that any adaptive change in the brain will minimize free energy this minimization could be over evolutionary time or milliseconds in fact the principle applies to any biological system that resists a tendency to disorder from single-cell organisms to social networks I could go on but that's his opening paragraph to introduce the concept of the free energy principle now I'll let you maybe comment on this so Sam what is the free energy principle yes so I should preface this by saying that I'm not either an advocate or an antagonist of the free energy principle I just noticed over the years that through many conversations with colleagues that there's widespread confusion about what exactly the free energy principle is and what it predicts and so that that was the motivation for me to enter into this fray and I should say that the free energy principle is closely connected to ideas that I've proposed and have been deeply

interested in and so I think the real question here is what is really special about the free-energy principle relative to other ideas that have been out there for a while and I can elaborate on that so in essence the free-energy principle is the idea that the brain minimizes surprise mathematically it's a little bit more complicated than that but that really gives you the gist of it now the term free energy is a little bit unintuitive for a lot of people because it actually comes from physics the idea was that you can decompose a complicated inference problem into inequality that expresses the relationship between what physicists called the partition function and what they called the variational of free energy and it turns out that the object that the physicists were studying is closely related to problems in machine learning and statistics where probabilistic inference problems are intractable and they can be made more tractable by turning them into an optimization problem under constraints so what does that mean and this this actually is perhaps usefully introduced by first talking about the Bayesian brain hypothesis and then talking about where the free energy principle enters into that so the Bayesian brain hypothesis states that the brain is confronted with ambiguous sensory evidence and it tries to interpret that sensory evidence by making inferences about the hidden causes that produced the sensory data so for example on our retina we get a two-dimensional projection of a three-dimensional world which means that the two-dimensional information is fundamentally ambiguous there's no unambiguous mapping from the 2d into the 3d nonetheless we're able to perceive three-dimensional structure in the way we do that is we combine the noisy or ambiguous sensory information with prior beliefs about the nature of the world so for example that objects have surfaces that are smooth and continuous right we can exploit that those prior beliefs to make inferences about the hidden structure of the world and that the Bayesian brain hypothesis is really that the brain is using Bayes rule to convert its prior beliefs into its posterior beliefs the posterior beliefs are the beliefs about hidden variables in the world given the sensory data or other kinds of information that the brain has received now that sounds very simple and if you just write down Bayes rule it's actually mathematically extremely simple and it's a direct consequence of the the axioms of probability theory the problem is that if the space of latent variables the hypothesis space is really large then implementing Bayes rule computationally is is totally intractable and part of the algorithmic challenge is to develop approximations of Bayes rule that can give approximately correct answers and that's where computer scientists and statisticians drew inspiration from solutions that were developed in in physics in fact a lot of algorithmic ideas in computer science particularly in the domain of probabilistic reasoning come from physics originally yeah and so the idea was that we can

convert Bayesian inference into an optimization problem and the way we do that is we say that there's some family of possible probability distributions and I'm going to try to pick one distribution in that family that gets as close as possible to the true posterior distribution right a good average to get us yeah well no it's not necessarily a first guess it's that I'm gonna actually try to search through this space of distributions to find the one that gets closest now the problem there is that if you just state it in that way it's just as intractable as the original problem because you can't compute the similarity between distributions unless you actually know both distributions in the whole name of the game is that we're trying to approximate the posterior distribution so the way that the key algorithmic trick that variational inference introduces is to put some constraints on the family so for example if you have a collection of hypotheses about the world like a bunch of different dimensions of the hypothesis space so for example you're looking at you're getting some 2-dimensional data you're trying to infer both the size and the weight of some object that you're looking at now in general size and weight may be correlated but you could make the approximation that I'm only going to consider approximations in which those two things are uncorrelated with each other so that that's sometimes called a factorization of the posterior and and it's related to what physicists called a mean field approximation so so it turns out that if you do that then you can you can simplify the inference problem and it no longer becomes intractable now what does this all have to do with the free-energy principle so the free energy principle is a form of variational inference that when we say that we're converting an inference problem into an optimization problem the thing that we're optimizing is the free energy and so so there's really no difference between variational inference and the free energy principle in general and in fact there's no difference between those things and the Bayesian brain hypothesis in general unless you put some constraints on the family of distributions the this variational family as they call it so once you start putting constraints then the solution to this optimization problem is no longer identical to the true posterior okay so that was a really nice introduction I think to the topic here so but you know you threw around a bunch of technical terms yeah and so that your paper is all about how you know putting different constraints on the free energy principle how it then matches up with the Bayesian brain hypothesis and so on I mean is this is this why the free energy principle is notoriously difficult to understand because it's a little slippery in that it's not well to say it's not overly constrained maybe in an understatement because it can go so many different ways it seems so plastic with the way that it can account for different processes I mean is that why it's so difficult to understand well I think there's a few different

factors here so one is that you're right that there there are a lot of degrees of freedom depending on how you implement the free energy principles so in in this context as I was saying a critical question is how do you parameterize the distribution the family of distributions that you're going to consider and then also how do you search through that space and actually solve the optimization problem efficiently yeah so it turns out that those different choices have important implications about things like predictive coding which we can talk about but I mean another issue here is that the papers on free energy principle are notoriously dense and long and hard to read and so I think that even quite technically literate people have trouble with that and so I was trying to clarify some of those issues yeah so I mean as you were talking about the Bayesian brain hypothesis I immediately jumped six analogies or something and just thought about the tabula rasa of you know potential tabula rasa of a baby being born let's say right and they come out and obviously they don't have an infinite number of distributions ready again to then be able to parse them down and constrain the actual problem of sensing mom's nipple you know whatever mm-hmm so and this actually gets to what we may speak about later in terms of what deep learning needs and the other paper but it's an interesting argument against a tabula rasa because if you come out as a tabula rasa you do have an overwhelming number of possibilities and you just stuck with too many options right of in other words your prior is infinite so how do you even come up with a posterior right right because how do you choose a prior at that point so so it's a really it's a really intriguing and comforting hypothesis the the and makes a lot of sense let's say common sense to me the way to approach how we learn and act yeah I mean I would just try to clarify that the free-energy principle is somewhat different from how we choose a prior it's certainly true that learning about the world is impossible without a prior there are theorems stating basically that if you're totally unconstrained and what you can learn then basically you'll never be able to generalize effectively but that's a little bit different from the free-energy principle which in my view is really about algorithmic approximations of probabilistic inference so even after you pin down your priors mm-hmm there's still the question of how do you actually compute with them how do you make inferences and that's where all the action is in frangie that's that's really what differentiates free energy approaches from just the Bayesian brain hypothesis for it largely it's really a subset of possible algorithmic solutions to Bayesian inference do you want to talk a bit about predictive coding I'm not sure that sir I mean I don't think it's not gonna be easy to convey the arguments passed but none of these things are easy really answer but but I would just make the point that well let's start let's start by summarizing what predictive coding

is all about so predictive coding has been a perhaps the most influential legacy of the free-energy principle for Neuroscience and the idea there is that there are ascending and descending or feed-forward and feedback pathways in the brain most of the time we're talking about cortex right and the feed-forward pathways are conveying prediction errors so discrepancies between observations and expectations at each level of the hierarchy and then the feedback pathways are conveying those expectations let's just contrast this with sort of a classic view of the feed-forward projections right classically you could think of well not classically I suppose I don't know what Classic is anymore but the feed-forward projections sort of building up a hierarchy of representation becoming more and more abstract would you say that's accurate yeah yeah that's one way to think about that it doesn't necessarily the pretty decoding doesn't necessarily exclude the possibility of certain increasing level of abstraction but it does make the strong claim which there is some experimental evidence for that the representation at a particular level of the hierarchy is suppressed that the feed-forward pathway is suppressed if that representation is anticipated so for example there were studies about a decade and a half ago using functional MRI showing that when you present a collection of line segments activity in primary visual cortex v1 is suppressed when those those line segments can be understood or interpreted as a shape as incoherent three-dimensional shape and concomitant activity in the lateral occipital complex which is known to be shape sensitive increases so there's a decrease in activity in v1 in an increase of activity and LOC and that was interpreted as arising from the this predictive coding framework where because the shape representation could predict the underlying line segments then the line segment representations become suppressed because now they're predictable gotcha now what does this have to do with the free energy principle well it turns out that you can derive this kind of architecture from the free energy principle but the catch is that you have to make a whole bunch of assumptions that have to do it a prokta variational family so how what space of distributions are you going to constrain the inference problem to you have to make assumptions about ways in which you're going to approximate the the free energy itself because it turns out that under under those chosen specify occasions you can no longer actually analytically compute the 300 so that anyway that the point is that there's a bunch of ingredients that have to work well together in order to produce predictive coding it's not a generic consequence of the free-energy principle as is sometimes conveyed yes oh so this is these are the things that you actually explore deeper in the paper so yeah yeah and of course the papers online and I think it's archived right yeah so I mean I'll link to it and pretty soon it'll

be in science I'm sure so so Sam what do you think like the Bayesian brain hypothesis and the free energy principle can both be seen as these over and these encompassing theories of the brain function in general right and how we aim to understand our minds and intelligence in general do you think that that will ever have like a single account of our brains and our minds well I think that the Bayesian brain hypothesis and free energy principle they are good candidates for some general principles but they're not really sufficient by themselves right that they're too general in some sense they're too powerful in order to act basically you could build agents that adhere to those principles and many agents in that set would be what we consider smart and many of them would be stupid wait isn't isn't that what we have already know just yeah right exactly so so but but my point is that if you want to pin things down sufficiently that you could say all right this kind of architecture these kinds of constraints these principles are only going to produce smart things or tend to primarily produce smart things then you need to make more assumptions than just that the brain is Bayesian or minimizes surprise I mean thinking about that do you think it's a better approach of course both approaches together is probably the best path but a better approach to go to start overly constrained or to start without enough constraints to start with something like the free energy principle and then and then start whittling away with constraints and and you know what I mean like what approaches is a favorable approach well there's a balance I know yeah I mean it's important to make a distinction between the engineering goals of building right artificial intelligence and the scientific goals of trying to parse what makes humans or animals intelligent obviously those things will converge in some ways but the scientific strategy is different in the sense that we want to design controlled experiments where we can isolate a particular hypothesis and test it that's different from engineering approaches which typically aim to construct something that does something useful and whether or not some particular component of it has been isolated is not that important and particularly in the era of building very large deep neural network architectures that don't have a lot of interpretability but can still accomplish certain kinds of tasks that it's you know the strategy the research strategy is different well speaking of building things maybe we could transition here and talk about deep learning in the state of AI and what it has what it needs what it's missing so this is your paper building people and this is in behavioral and brain sciences mmm-hmm and this is a thorough paper because it's in behavioral and brain sciences and made even more thorough by the the format so just I mean people know this but but there's a target article which you guys wrote and then there's a bunch of reviews and then you get to rebut the reviews and this is all in like a single document and it

costs about a three and a half - if you really want to print the whole thing out you know so I've always liked that format but I've not written one like that mm-hmm and it's a lot of work just to get reviews I'm one of my papers under is under review right now and you know just with with three to four peer reviewers it's a lot of work just to rebut that and send it back but it seems like a lot more work for you guys that I mean how did what was that experience like thing is if it was written last year you probably did the work twelve years ago you know so yeah yeah no it was it was a lot of work but it was fun to write um you know is it it was great writing that paper with those co-authors because it was these were ideas that we had talked about for a long time and it finally forced us to really coalesce them yeah okay sorry I had Melanie Mitchell on the show a few weeks ago and she's a complexity expert and she's been in the field of AI for a long time and she had written this op-ed in the New York Times yeah I read that oh yeah okay so so here's just a quote from like her approach to this today's AI systems sorely lack the essence of human intelligence understanding the situation's we experience being able to grasp their meaning the mathematician and philosopher Gian Carlo rota famously asked I wonder whether or when AI will ever crash the barrier of meaning and so there's this phrase the barrier of meaning and and this paper that you co-authored takes that sort of broad statement and breaks it down into cognitive components that could help bridge that gap so it's really an initial road map of sorts and has seemingly served as such which is pretty must be pretty rewarding for you to have seen so I mean maybe we can frame this this way and maybe we can just start off talking about how machines think and learn currently and then maybe bring in how humans think and learn what we know about that and then talk about them together so so the premise of the paper is like yeah we're in this we've done great work with AI and deep learning has done amazing things but it's still not human-like and if we want to make it human-like we need to build in these other components these higher-level components we either build them into the deep nets or however you're going to perform the AI build these core ingredients in do you want to just talk for a moment about on the current state of how machines think yeah well it's it's quite hard to make a really general characterization of how machines think because there's a hundred papers posted on archive every day on machine learning it's a rapidly moving target and our paper was really focused on kind of general patterns right so it's not the case that our critique yeah Gaius every machine learning paper that was ever written but these are these are the base that have been going on for a long time and we discussed this history a little bit in our paper I mean since the 70s and 80s but by the way it's a beautiful paper I really enjoyed it I bet you I bet you've heard from a lot of

people the same as well thank you yes so people have been going back and forth about the necessary ingredients for intelligence for a long time and I sympathize with computer scientists in some ways because there's this feeling of moving goalposts like you're there's a time when people said you know when a computer can win at chess that's when we know it's really intelligent then if it wins I go then it's really intelligent and it's a sort of moving goalpost and you know I'm sure some people wonder like when are you going to be satisfied and I'm not really a big believer in having a kind of single test mm-hmm of that sort because I just think that human telogen s-- is too multifaceted to really be boiled down in that way but that being said I do think that it's important for us to think about what we've learned from study of cognitive science and what that tells us about the nature of human intelligence and the limitations of current AI approaches so with that premise in mind that I can kind of make some broad stroke characterizations the basic thrust of modern machine learning approaches particularly deep learning approaches has been to construct systems that are extremely expressive in the sense that they can learn lots and lots of different things so we could think about this as a kind of function approximation problem like you get you have a function that maps images to object categories and that and that's that's one way to think about an object recognition system or you have a translation system that's mapping text in one language to text in another language so these are all potentially really complicated functions and the the general approach has been to build very complicated neural network architectures now I should say they're not conceptually complicated you can like basically draw a picture of them and it's pretty straightforward you have a bunch of neurons that are connected to each other in layers but they're complicated in the sense that they're very expressive they have lots of they have lots of degrees of freedom possibly millions of parameters that then get optimized by a learning algorithm and so you could you could take the same the same Network same naive network or whatever in training on a bunch of different types of domains right so you could yeah use this thing to Train it on recognizing objects you could use this same network to train it on language or you know something within the real there's a lot of possibilities yeah to Train it yeah so it's just to maybe put this in perspective before the heyday of deep learning the most successful approaches used collections of hand engineered features so people would put a lot of thought into defining various kinds of image or natural language features depending on the domain and then they would assumed a fairly people supervised learning algorithm that operated on those features so for example you might assume that there you take it this collection of features you map them through a linear function you you map them through a

linear classifier and then you basically train the weights of that linear classifier mm-hmm so that's a very low complexity solution in the sense that the space of possible functions that can be learned is very is relatively small right so you can only learn linear functions of this feature set contrast that with a deep neural network where all the different parameters and all the layers are being optimized to solve this classification problem that's what's commonly called end to end training and now you have millions of parameters in the function space is enormous because now you don't necessarily you're not necessarily restricted anymore to linear functions you couldn't learn linear functions but you could learn all sorts of nonlinear functions so you might think ok well that's great now we can learn more functions right but that the catch is that the more complex your function class is the more data you need to effectively learn it in the sense that if you only have a small amount of data then you could learn you could learn to model those data very well right if you have a very large function class you can you can fit the training data really well but the problem is that there may be noise in that training data or the train data might not be representative sufficiently of all the test cases you might encounter and so your your classifier is going to fail to generalize and and there's a whole literature that rigorously analyzes this from from a statistical learning perspective it's called the study of sample complexity how much data do you need to effectively learn to effectively generalize and in in general more complex models need more data and that's why actually the in some sense the deep learning revolution has been just as much about Big Data revolution in the sense that we've constructed bigger and bigger data sets to feed the insatiable sample complexity needs of these learning algorithms plus the computational power to do so yes absolutely and now it's the other thing so yeah big computation is a big contributor in some sense the deep learning architectures we're using today are pretty similar to the ones that we had in the 90s but they work better because we have a lot more data and a lot more computing power that being said it is still somewhat of a mystery to learning theorists why these things work because you know the it at first blush the learning theories would say that you would need even more data than we have to effectively generalize but learning theorists are trying to crack open that nut and that it's probably not worth getting into that right now but but but the point here is that the approach today is about building really flexible function approximator x' now that doesn't sound so sexy from the perspective of artificial intelligence to boil them down to nonlinear function approximation but that's really what it's all about now as I was saying the architecture of these networks is really flexible but it also means that that that that very flexibility allows the system to learn things

that we might think of as not being the right answer hmm so it's going to pick up all sorts of patterns but it might fail to learn the same kinds of things that humans learn and and they might not learn those things in the same way so let me try to elaborate on what I mean by that so I was talking about complexity in terms of the number of parameters and the size of the function class but another way to restrict the effective complexity of a machine learning system is to have an inductive bias so it says that an inductive bias I said all right I couldn't Theory learn any function in this function class but I have a preference a priori for certain functions over others so I think that some functions might be more plausible than others and I can formulate that in the language of probability theory and that shapes the sample complexity arguments because now even if I have a really large function class if I have very strong inductive biases that can kind of compensate for the four the need to experience a lot of data that inductive biases basically say that some things are going to be easier to learn with less data and other things are going to be harder to learn and will require more data and is that how does that compare to having a stronger priors those are rights right so inductive bias these are priors when you can you can construct depending on what you're you're kind of formal framework is you can talk about inductive biases in a in a non Bayesian way but I often think it's helpful to think about prior distributions as a particular way of representing inductive bias if I say okay so so now so let me try to loop this back to human cognition we think that humans can learn from small amounts of data precisely because they have very strong inductive biases hmm and they also have inductive biases that are relatively abstract which allows them to generalize their knowledge very widely so let me just talk about some of those inductive biases actually let before before I talk about inductive biases let me just try to make vivid why you need an inductive bias at all there's a great book by Eric Baum called what is thought and he has the following example so imagine I'm training a chair classifier so I'm giving it images and it's telling me chair or not chair all right so I give it some finite number of these training images and it learns to perfectly classify to perfectly discriminate chairs from non chairs all right now I'm going to give it a new chair and I wanted to generalize to that now let's imagine that it had no inductive bias okay the implication of having no inductive bias is that in fact this classifier which could perfectly discriminate between chairs and non chairs in the training set has no basis of generalizing to this new image that in fact it will be completely uncertain about it would have no basis of knowing whether this new image is a chair or non chair and the reason is because in a continuous function space there are an infinite number of functions that separate the chairs from the non chairs and that means that there's an infinite

number of functions that would perfectly separate chairs from non chairs and and classify this new image as a chair and then there's also an infinite number of functions that perfectly separate chairs from non chairs and classify this new image as a non chair okay and what it means to have no inductive bias is that you have no preference about which of those functions is the correct function so generalization depends crucially on having some inductive bias now neural networks do have inductive biases the architecture of the network the parameterization and so on those are all at forms of inductive biases but they're just relatively weak inductive bias right now we think that humans have strong inductive biases they haven't strong inductive biases about things like causality about objects about relations between objects about compositionality of knowledge so that the idea that you can build complex representations out of simpler ones and we have strong inductive biases about I in different domains that take the forms of something like an intuitive theory like an intuitive theory of psychology that lets us make inferences about other people's minds or an intuitive theory of physics that allows us to make inferences about physical interactions in the world anyway I'll stop there and let you look well no that's that's okay because you're sort of rattling off the the ingredients that are proposed in this in this paper so and you just got there naturally which is a you know a beautiful thing yeah so so I was gonna ask about how humans learn right and in in comparison which is where you've really just brought us to anyway but before we even get into just a few maybe of the examples of these things that you just listed it's an interesting conundrum whether we understand how humans learn well enough to then so the whole goal is to build systems that learn and think like humans but do we learn do we know how humans learn and think well enough to even begin doing that and so these are things you know like when you say terms like intuitive physics oh of course well let's just maybe even put that question on hold unless you want to answer it and then we can kind of step through a few of the ingredients well no I I mean I don't think that we know enough about human cognition that we can just implement human intelligence in a computational system it's more just that we have some general hints and also some specific computational models about how humans think and those are in interesting ways different from a lot of the prevailing deep learning architectures now I will say that even at the time that we were writing that behavioral and brain sciences article people were already starting to realize this so for example in the domain of intuitive physics there was a lot of interest in building machine learning systems that that solved intuitive physics problems and if you build them in a kind of naive sort of off-the-shelf way they didn't work very well right and they only started to work really well when people started

building in constraints and and I would argue that those a lot of the constraints that worked well were precisely the constraints that we think people have so for example deepmind and other groups have developed models in which there are primitives that correspond to objects and there are functional interactions between the objects so it's a relational model that's object-oriented and then you use deep learning to learn the parameters of those functional interactions and that works much better for for modeling intuitive physics so I would argue that that's kind of harnessing the strengths of deep learning which are really about flexible function approximation with the kinds of inductive biases that we think humans have okay so let's just step through these so there are five core ingredients is that the latest count I mean this is the sort of the count in the in the paper huh yeah so so one of them is this intuitive physics that you're just talking about and this is the idea of so so this falls under the rubric of when we're really young we're kind of born with and have really early on this developmental software right so that's sort of ingrained or an inductive bias right so so intuitive physics is the idea that we understand that one cue ball hits when one pool ball hits the other billiard ball we understand that there's a collision there these two things are solid and one affects the other yeah well it's a lot more than that probably but well yeah that's one aspect of it yeah that's one aspect of it so you know the things are solid not illiquid and and yeah and so on and so on so so this is one thing that is sort of missing in deep learning networks that you were just describing that's and there's being there's work being done on all of this like I said it's uh it's this is the nice roadmap that people have seemingly been following actually so another one of those that falls under the rubric of developmental software is the intuitive psychology like you were just that you that you mentioned do you wanna just describe what that is briefly yeah so so maybe just first for those people who are not familiar with the notion of an intuitive theory the reason that they're called intuitive theories is to contrast them with scientific theories yeah and the argument is that intuitive theories and scientific theories shared a lot of the same structural characteristics but obviously they differ in content in the sense that you know intuitive psychology doesn't have access to you know functional MRI and response time studies and so on right but but we still can build a theory like representation of another person's mind for example we can reason about another person's beliefs and desires and combine our beliefs our knowledge of those beliefs and desires to for example make predictions about what that person is going to do mm-hmm another example would be you know when kids are shown kids can discriminate between whether an acting agent is is helping another agent or hindering the other agent and so the yeah there are these

psychological aspects of other agents that can easily be inferred with this sort of developmental startup software yeah okay and then the rest of the other three ingredients kind of fall under what you describe as model building elements and one of them is this company and you've listed all these already if you've mentioned these in passing so one is cop isn't compositionality mm-hm and this is basically where you can build new things from simpler parts mm-hmm and so deep networks and let's just run through these real quick and then we can kind of talk about them more broadly maybe the deep networks are not great at this another is learning to learn or meta learning and so there's actually a lot to talk about with this probably and and causality is the last the fifth ingredient and building causality and I talked about this last episode a little bit with Conrad hurting and you want a modern ESCO his wife because they just wrote a paper on how to infer causality in quasi-experiments so it's a little bit different okay so like I said this has been a road map and it's like the paper was written and then learned meta-learning work immediately came out of deep mind and then I don't know if you recently I didn't get a chance to read the paper but there's this recent paper out of deep mind as well that is using the meta reinforcement there meta reinforcement learning framework to build in causality tool it was written by my student oh is that right well I should have internship that at deep mind okay very good yes it's funny because in the paper and one of the rebuttals from the people at deep mind they're like maybe we are on a converging path here and so I suppose you are I know we're definitely there's definitely points of convergence there not not completely but I mean I talked to those people on a regular basis so it's not we're not like you know in conflict or anything oh of course not we're all in going for the same goal anyway yeah so so I don't know if you want to if any of those that you want to really elaborate on as something that you know I'm kind of curious your take on how the paper has continued to be received over time and and how people you know if you've seen a lot of progress in all of these domains or what's really still lacking etc yeah I think I think that there's been a ton of progress I think that as I was saying before many people have begun to acknowledge that you need to build in some constraints to get these systems to work appropriately now we're certainly not there yet in a lot of practical areas of application you know for example we we still I think can't get systems to make reasonable captions for images it's just the fact of matters they just don't really understand what's going on in the image the barrier of meeting but they make reasonable captions you know five out of seven times maybe now is that a good I don't know I mean nice okay yeah yeah anyway but the point is just that these systems can sometimes fool us into thinking that they know more than they really do and yeah so so I'm

encouraged by things like the paper you mentioned meta learning for causal inference I think that those kinds of approaches are definitely on the right track I don't know to what extent the particular basically the thing that I wonder about is to what extent building exploring in the space of kind of different neural network contraptions putting in different kinds of widgets and so on is going to get us to intelligence and sometimes it feels a little bit like building a ladder to the moon doing more the same that didn't get us there initially and yeah well it's just it's just I mean certainly there's no doubt that there's been a huge amount of progress but it also you I get the sense reading these papers that we're kind of swirling around a local optimum and every once in a while we kind of pop out of that and go a little bit farther but the developments don't seem of the sort that are really getting a significantly closer to human intelligence so I it's just um to give one example I think about alphago you know alphago is an amazing accomplishment look I love and everyone has to always preface it with how amazing it is and there yeah yeah yeah well it's an amazing accomplishment there's no doubt about that but just think about what happens at the end of a go game you shut it down because público can only play go and that's in some sense true of almost all the models that we have now and even even with the space of let's say video games so we have algorithms that can play video games but you can make changes to those games that would be completely trivial to a human player and yet would be catastrophic to these algorithms yeah so we're really far away I think in the sense of capturing that human-like flexibility and I and I think it's because we haven't fully embraced the principles of causality and compositionality and intuitive theories and so on I think I think I guess my general impression is that we have this really useful tool that we can build these gadgets out of neural networks that do things and you if you can feed it and updaters they can you know classify images or recognize speech but it's tempting to kind of over apply those things as a substitute for really thinking hard about the structure of a problem but it'd be fair to say that that it's a sort of a change in quantity when you're adding these sorts of widgets etc to the deep learning networks and not quality yeah yeah I think that most of the advances that we see on a day to day basis are about like you know getting a few more decimal points score on some benchmark data set but you're not really you I think the real contribution of cognitive science is to get people out of the thinking in terms of get people move people away from thinking in terms of these benchmarks and and or rather thinking about thinking more broadly about what the benchmarks really are what what is it that you want the system to do it's not a just about playing go or playing video games or recognizing objects there's some set of competences there that transcend those individual tasks and if you

only kind of play to win those tasks then you're going to be missing something so we need to figure out where the hell we actually want the goal posts and pour concrete and then make them stick there huh yeah yeah yeah so I asked this to a lot of my guests so I'll ask you as well is emulating the human mind is that you know is emulating human thought and learning is that the only way to get there you think too to general AI to something that we would satisfy that would satisfy our notion of what generally is well can I tell you what you're gonna say before you say it yeah go for it you're gonna say well it's the only example that we have it's the best example it's the most complicated you know the brains most complicated thing in the universe okay now go ahead and repeat what I said and I'll be saying I think it depends on what you mean by general intelligence because it could potentially be a tautology if we think that what we mean by general intelligence is whatever we think humans do then that's only going to be satisfied when we build machines that think like humans but we might not right maybe we'll eventually come to some way of thinking about intelligence that's really different from our own it's just it's very hard to get to that point because inevitably we project our own notions of intelligence onto our machines of course but I mean one way to approach this potentially is from you know the computational level so we can always talk about David Marr and the conformational highest in level and where where would you start like the free energy principle for instance I think starts with existence itself right as the you know the fact that we exist and if you start in that level well then what's the for living things and what's the algorithm maybe that's is that evolution and you know I don't know is there is there a way to come at it from a really high computational level where then the algorithms could differ or you know for instance and then the implementation level below that could differ as well to get to that same whatever the actual computational level is of living in existence this is getting a little too deep I'm sure for the for the podcast but well we I think that's absolutely right I mean there's some sense in which like at the very highest level it's to like do well at life right just survive and reproduce the imperatives of natural selection and so on but that as we talked about before is not very constraining on theories of intelligence at least in its most general form yes so we need to make more specific assumptions but but just to make this a little bit more concrete so even if I said all right an intelligent system has to reason according to Bayes rule then there's they're going to be algorithmic constraints to grapple with in intractability of that in any large domain and it's perfectly conceivable that a machine we could design machine learning algorithms that Excel whatever algorithms humans are using because there's no reason to think that evolution is necessarily selected the best possible algorithm to solve these problems there

is no best best is just what is existing at the time I suppose right right why I mean yes so I'm saying within whatever particular niche you're looking at you could ask like what is the best way to solve this problem and humans might not have achieved that so for example people might engage in some kind of tree search to solve planning problems or they might engage in some kind of like variational optimization to solve inference problems but the particular way in which they do that might not actually be the optimal way and we could we could design machine learning algorithms that do better than humans now they're still trying to their design to achieve the same goals that humans want to achieve and that the overarching architecture of that system is similar but it's doing it differently right so that's that's a concrete way in which it wouldn't even make sense at that point to demand that the machines adhere slavishly to it would make no sense what humans are doing right yeah yeah so I I mean I the reason you know I told you what you're going to say I mean I was a joke obviously but yeah so many people have said that you know they can't imagine another way and my intuition is that wow you know if you ask a hundred experts and they all say no way we have to do it this way it must be the other way you know you know I can't I can't imagine another way but I can imagine right there being another way in the same way that I was very struck by I think this has very deep implications for how we think about ourselves and our own thought process so I attended a dinner once and I talked to this geologist who was working on earthquake prediction and he made the very vociferous point that he felt that the classical education and geology had ruined generations of geologists come because the classical models simply could not accurately predict earthquakes like tectonic plate shift models or this is what he said I'm not a domain occurred and he didn't really go into the details of why that was yeah yeah that you could have sound geophysical principles that were not actually useful at making predictions and he argued that this whole thing was blown wide open when people started to use deep learning to predict earthquakes and he felt that whatever it is that we thought we knew about geology you know even if it was right maybe there are more correct theories of geology that are expressed in some inscrutable language that the deep learning system can discover yeah and eludes us and I I have a hard time swallowing that but at the same time you know you have to acknowledge the possibility that it's our hubris as humans that convinces us that the nature of the universe has to be expressible in the language that that we have at our disposal right now in the scientific language in current use I mean this was this is something that historians of science have known for a long time that our theories are strongly latched to our description languages and even our basic observations can't be described in the theory

independent way when our theories change our observations change we actually see things differently and so so that's reason for humility in terms of privilege in our notions of intelligence right right well in that same vein I had John Krakauer on the show a few weeks ago and and we were talking about different levels of explanation and you know we talked about reductionism and in the proper level of understanding phenomena and there are lots of like high cognitive functions that it may not make sense to try to understand on the level of neurons and neural circuits right but instead on the level of some behavioral function and and so on and the like the core ingredients that you guys mentioned in your paper confer like these advantages right they make humans so awesome in various ways and I'm wondering you know could there be disadvantages too and they'll have to be right that that's the yin and yang is the trade I yeah so that's what I was thinking too is like you know let's say it is possible you know we can map some of these these higher level cognitive functions to the level of neural circuits you know are we humans but you know because we need to reduce complexity using things like compositionality and and intuitive physics are we incapable of conceiving how that mapping could occur whereas like a deep learning network or some sort of AI system without these human-like skills quote-unquote or limits might by brute force figure it out in a way that we could eventually be satisfied with that kind of explanation you know does that does that make sense yeah I so i wouldn't say that people are incapable of doing this since historically we've been very capable of constructing various kinds of cognitive prosthesis that allow us to understand things that seemed mystifying before that being said it's certainly a logical possibility that neural networks could discover these theories maybe more easily than us because they have different inductive biases than us I certainly think it's within the realm of possibility to think about neural network scientists that are as different from us as we are from you know dogs and cats now not as but but that being said I think my experience in studying these systems is that in some sense there's always less than meets the eye that like that we have this strong inclination to kind of anthropomorphize these systems and attribute to them psychological competences that they lack and when you start to really test these rigorously they fall short I mean this is very clear in the domain for example of natural language where most natural language processing systems simply don't recover the same kind of semantic understanding of the world that humans have it's not that hard to demonstrate this so so I just think that while it's a logical possibility I think we need to be doing more than just training neural networks on imagenet before like you know data set they eat even data sets on reasoning from natural language I think are too impoverished but see isn't it it's a it's really a

privilege to be able to be at the point where we can say that and you know that there have been these advances right that are amazing now we can say it's not enough that's great I think that's a place to be in mmm-hmm so I can imagine like lots of different scenarios for this path toward puzzly what we would could call general AI and reading your paper brought just one to me is this one scenario and as soon as you write this in a science fiction book it means it's gonna be completely wrong but so you know I can imagine and you know the way I the way I sort of envision it is what we're building toward is an AI system that comes off the shelf out of the box and it's ready to solve new problems and learn new things efficiently and and and have some of these built-in ingredients right that that are proposed in the paper mm-hmm I can also imagine a scenario though where we the service and in between phase where we have to build these machines that then we set out into the world like with little embodied cognitive architectures and then we have to wait for them to gather enough different kinds of data and experience that we can't build into them mm-hmm you know do you think that's a possibility that will go through this phase where we're you know taking these these BOTS these robots and stuff and then letting them be for a while and then we can use them for instance then then their general domain general enough that we can actually use them or you know can you imagine that kind of scenario it's a ridiculous question okay yeah I mean I don't see why not although of course there's all sorts of ethical and practical issues sure that loose robots in the world in depends what kind of constraints we put on them and what kind of challenges you know do they have to enter the workforce and work for there oh definitely definitely yes for me anyway yeah so I came up through neurophysiology in my my graduate work and my postdoctoral work and this podcast I made has the tagline we're neuroscience and AI converge but I always have thought of and and I'm only starting to not think of it this way I've always thought of cognitive science as a part of neuroscience now that neuroscience sort of subsumes psychology even you know mm-hmm it seems to depend on whose definition you like of these things but so this paper was written in a real from a real cognitive science historical perspective and modern-day sense of of cognitive science do you think that we're in a cognitive science resurgence sort of over over neuroscience so there was a lot of hype in neuroscience right we're gonna figure all this out and now cognate is cognitive science coming back now and saying whoa whoa you know this is the way to go are we in this resurgence to understand how how to understand the brain and how to programme machines that think like brains it depends who you ask I mean there's certainly still people who argue very strenuously that neuroscience is really important for building intelligent machines I just I don't think either

for cognitive science or for neuroscience you know psychology or neuroscience they're both in some sense very dependent on our computational notions we need ideas from machine learning and other disciplines technical formal ideas to interpret both psychological and neural data I don't think it's the case that we've gotten a lot of mileage out of just looking at those data and from that extracting algorithms that we can plug into machine learning systems I just don't think that's how it really works people have different arguments about this of course yeah so let's kind of transition I just have a few general kind of broad questions for you and then we'll then I'll let you go okay I will see how you perform on this one so in the spirit of human compositional skills and our ability to produce infinite things from a finite set of things mm-hmm can you say I'm uh Turei grammatically correct English sentence right now that you think nobody has ever said or written sure yeah I don't think that is I mean you could do any variant of Chomsky's classic sentence colorless green ideas sleep furiously you could have them sleep quiet sleep professionally effervescent Li yeah colorfully they dead verbs it's the adverbs that allow them to do this yeah yeah I think it's pretty straightforward to do that okay um I could you know I could say the sentence I don't believe that Chomsky said that colorless green ideas sleep furiously I bet that's been said but we'll see Oh someone can google it okay so often when we're stuck and need to figure out the solution to some sort of problem we need to step away from the problem right you can't just like you know you're at your desk you're trying to figure this thing out but it's only later when you step away that you know the solution comes to you and often like immersing ourselves in a like long-form complicated novel for instance can lead to more creative ideas than like reading a summary of that novel right presumably because we need to sort of steep in that mind frame for longer not the things bubble in or in our minds do you think that there is something about time scales in this I thought you might be going to this when you're talking about your dinner with the with the geologist is there something about time scales and dynamics and processing time that is important for how we think and learn and how maybe machines might need to think and learn so so thinking of you know the idea is can like the the the geo geological formations right they're moving and they're they're a dynamical system could we if you stepped back on a really slow timescale with those look intelligent could we make the case that those looked intelligent right or is there something special about the millisecond processing times of our brain that gives rise to intelligence and that we need to consider when building machines I don't think so I mean it's not entirely clear whether you mean that the time skills refer here to the time skills of computational processing versus the time scales of events unfolding in our

world I could clearly we can reason about events that happen on it vastly different time scales even though the time scale of our neural processing is fixed I didn't say they're gonna be good questions yeah constraint yeah I mean I think that there's something important about time scale but I'm not sure what it is I mean it is an intriguing question why thinking over long periods of time would help you what what is a one idea that you seemingly you have time to do all your ideas but what's one idea that you either you know don't have the resources or time to tackle that you wish someone else might pursue yeah well a lot of my questions about the brain really need to be answered in animal models and although I do human experiments in my lab I don't do animal experiments and so typically I try to farm out those animal experiments to other labs that do that kind of research and that's been no no pun intended yeah that's been very fruitful for me to answer questions for example about the nature of dopamine processing so there's lots of questions like that I mean often the kinds of experiments that we do we often try to design them in such a way that both humans and animals could do those tasks and then eventually eventually try to port them over to animals and there's some yeah you know process of convincing an animal experimentalist to take on that challenge well they're busy you know other things going on well it's not just they're busy I think they're also they're conservative understandably because it takes a long time as you know to Train animal that's what I mean they're busy yeah steeped in their two-year other other experiment if you could bet on a single discipline or sub discipline that will that will break open our understanding of the human mind what would it be you could say cognitive science physics astrology etc well I feel like the longer that I do this stuff it just seems that the more everything kind of converges that we already talked about how physicists came up with this idea of free energy that then got picked up by computer scientists to develop efficient inference algorithms that then got picked up by neuroscientists to think about the brain yeah I mean these all all these things are intertwined they're they're deep both formal and metaphorical relationships between all these concepts and so I really like what study of intelligence is that in a nutshell right you need the intelligence is drawing upon all of these different disciplines and ways of thinking so in that vein keeping one's mind as broad as possible do you have a advice that you would give to you know say a graduate student or an undergraduate interested in these topics and just wanting to start out in the arena like what how to how to move forward yeah I think it's hard for students to follow the path of greater resistance because they look around and they see what's happening what's successful what's not successful and it's natural for people to gravitate towards the things that they perceive as

being more successful but the paradox is that to be really successful to really make a difference to change the way people are thinking to change the course of scientific discovery you have to swim against the current you have to pursue ideas that are not popular that people might not agree with that but that might not really make sense at first you have to befriend crackpots and read poetry and you know take long walks and do all the things that aren't ostensibly the right things to do the productive things the good uses of time right so so we have to be really attentive to the unappreciated importance of uselessness well that's a good quote i tempted in there but I'll I have one more question for you here so yeah well AI help us understand consciousness what it is how it works etc yeah I'm not sure should it ended with the last one I I don't feel like I can really give a good answer to that because to me most of the things that we do that are intelligent are not necessarily conscious or what what's the conscious part of them is just kind of the surface the perceptible surface but most of the interesting computation is going on under the hood and that's why the study of psychology has moved away from just asking people what they're thinking is demonstrably people don't really know what they're thinking mm-hmm so in my view consciousness is a little overrated damn alright well I know what I'm thinking I'm feeling grateful and thinking I'm glad to have you on the show so Sam where where should people go to learn more about you well they can look at my lab web page and if they're interested in reading my papers that's where they all are very good that'll also be linked to in the show notes so so thanks for your time Sam and continued good luck to you yeah thanks so much Paul take care brain inspired is a production of me you can support the show through patreon for a trifling 2 or 4 dollars per month go to [patreon.com/scishow](https://www.patreon.com/scishow) and inspired co find the red patreon button there your contribution will help keep this show going without any annoying advertisements like you hear on other shows to get in touch with me email Paul at [braininspired Co](mailto:braininspiredco@gmail.com) the music you hear is by the new year find them at the new year net thank you for your support see you next time [Music]